

Time-frequency Based EEG Motor Imagery Signal Classification with Deep Learning Networks

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Abstract—In this research, a wavelet transform-based feature extraction approach with time-frequency analysis is proposed for motor imaginary EEG signal classification. The proposed approach selects specific channels such as *C3* and *C4* to identify event-related synchronization (ERS) or event-related desynchronization (ERD) phenomenon to filter out the artifacts and noisy data from signals. As EEG dataset is noisy and size of the dataset reduces after filtering, the proposed approach adopts multi-scale analysis ability of wavelet transform to utilize small input. It allows to extract features from the dataset and generate input images for training the models. Considering abstraction ability of Convolutional Neural Network (CNN), deep CNN with two convolutional layers, and VGGnet with six convolutional layers are employed. The model performance is evaluated in terms of accuracy, loss, and epochs. The proposed approach is applied to EEG dataset III from BCI competition II. The primary results show that VGGnet performs better than deep CNN with respect to training loss and training accuracy.

Index Terms—Electroencephalogram (EEG), Event-related Synchronization (ERS), Event-related Desynchronization (ERD), Brain-Computer Interface (BCI), Motor Imagery (MI), Convolutional Neural Network (CNN).

I. INTRODUCTION

The active part of human brain changes its pattern when human imagines that s/he is performing the limb movements. In such a case, contralateral of the human brain sensorimotor cortex show phenomena that are called Event-related Desynchronization (ERD) and Event-related Synchronization (ERS) [1]. The existing challenges in motor imagery classification for EEG signals permits Deep Learning (DL) as a progressive tool. In this perspective, Artificial Neural Networks (ANNs) with different architectures are widely employed for predicting and classifying the motor imagery signals [2, 3]. The development of other deep learning models causes a research gap to justify their performance in motor imagery classification from EEG signals [4, 5].

In order to develop a classification model for motor imagery EEG signal, a wavelet transform-based feature extraction approach is proposed. The proposed approach takes into account the two-dimensional time-frequency analysis to obtain detailed information about time-frequency and electrode position as features for classifying the motor imaginary EEG signals. Moreover, the approach selects specific channels such as *C3*

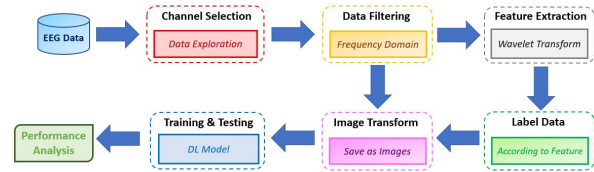


Fig. 1: Flowchart for the motor imaginary signal classification.

and *C4* to identify the ERS or ERD phenomenon to filter out the artifacts and noisy data from signals. The EEG dataset is noisy and after filtering, the size of the dataset reduces. That is why the proposed approach adopts multi-scale analysis ability of wavelet transform to utilize weak (small amplitude) signal data point for extracting features and generating input images in order to train the models. As wavelet transform does not follow the window selection method in Short-time Fourier Transform (STFT) method, the abstraction ability of deep CNN models (i.e. deep CNN with two convolutional layers and VGGnet with six convolutional layers) are used for training the models from the dataset. The model performance is measured in terms of model accuracy, loss, and epochs. The proposed approach is tested and evaluated on EEG dataset III from BCI competition II.

The rest of this paper is presented as follows. Section II introduces briefly the proposed approach of motor imaginary signal classification. The applied deep learning models are discussed in Section III. The performance validation and result analysis for different deep learning models are shown in Section IV. Finally, a summarizing comment with future direction of this work is given in Section V.

II. PROPOSED APPROACH

The motor imagery tasks vary from person to person and their characteristics reflect in different frequency bands of human brain extracted EEG signals. Most of the motor imagery tasks can be captured as ERD or ERS pattern in the EEG frequency range of 8 – 30 Hz. To extract the correct frequency to identify individual task is challenging within a narrow frequency range. The steps that are used to classify the motor imaginary signal is shown in Fig. 1.

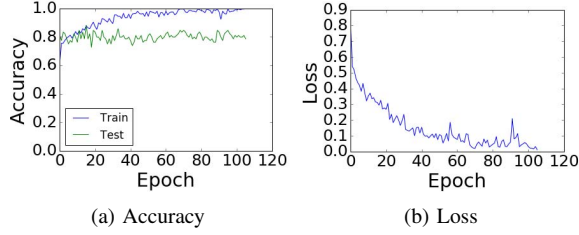


Fig. 2: The performance of Deep CNN.

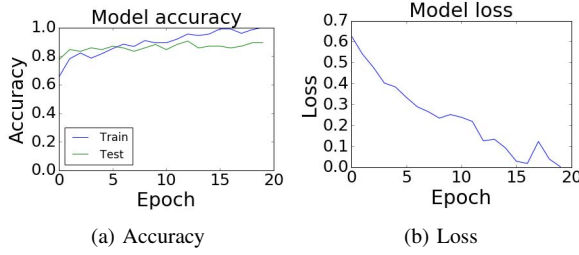


Fig. 3: The performance of VGGNet.

III. DEEP LEARNING MODELS

The evolution of mainstream network in deep learning allows researchers to capture the hidden relation of the data with efficacy. The mature networks are not performing well for a small dataset like BCI datasets due to lack of sufficient data points. Moreover, training the network with less possibility tends to be over-fitted and challenging to oversee. In this research, the signal information is converted into image data for applying some of the mature networks with reduced number of layers. Hence, deep CNN with two convolutional layers and VGGnet with six convolutional layers are employed on the dataset.

IV. PERFORMANCE VALIDATION AND RESULT ANALYSIS

Following the proposed approach discussed in Section II, the parameters that are associated with experimental analysis are summarized in Table I.

TABLE I: Parameters for Experimental Analysis

Topic	Value
Validation split	30%
Batch size	5
Epochs	500
Filters in WT	"Butterfly Filter"
Input/converted image shape	$64 \times 64 \times 3$

Since the EEG dataset III from BCI competition II contains data for left or right motor imagery signals, the datasets are trained with two different networks to compare the performance in terms of accuracy and epoch number for reaching the maximum training accuracy. VGGNet outperforms deep CNN in terms of accuracy as the maximum, minimum, and average accuracy of VGGNet are higher than the deep CNN (shown in

Fig. 2 and Fig. 3). Moreover, the model loss of VGGNet is less than the deep CNN. Although the average model loss of these two networks is very close to each other, the minimum and maximum model loss of VGGNet is less than the deep CNN network. In addition, the training epoch number of VGGNet is less than the deep CNN as deep CNN takes a higher number of epochs (110) for reaching the maximum accuracy (0.904) during the training comparative to the VGGNet that takes 20 epochs to reach the maximum accuracy 0.904.

V. CONCLUSIONS AND FUTURE WORK

In this study, a motor imaginary EEG signal classification approach was proposed that followed specific channel selection to identify the ERS or ERD phenomenon and later performed a filtering approach to remove the artifacts and noisy data from the signals. The wavelet transform was introduced in this approach considering its multi-scale analysis ability without STFT window size to extract the features from the dataset and generate the input images for training the models. The abstraction ability of deep CNN allowed a new means of motor imaginary EEG signal classification. Model accuracy, loss, and epochs were taken into account as performance metrics for classifying the motor imagery EEG signals. The results showed that although the accuracy of deep CNN and VGGNet is almost close, the VGGNet takes minimum epochs to minimize the training loss and maximize the training accuracy. The future direction of this research is to apply the model with other performance metrics in a real-time system to analyze the performance deviation.

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